Jaron Arbet, Ph.D.

Assistant Research Professor - Biostatistician

City of Hope - Beckman Research Institute - Department of Population Sciences

jaron.arbet@gmail.com | jarbet.github.io | Google Scholar | Github

Research Interests

- Machine learning and predictive modeling
- High-dimensional “Big Data”
- Variable selection
- Multi-omics cancer data integration
- Robust nonparametric statistics

Work Experience

Assistant Research Professor - Biostatistician

Dec 2025 - Present

City of Hope - Beckman Research Institute - Department of Population Sciences, Duarte, CA

- Studying the impact of exercise on risk of cancer diagnosis, and for improving health outcomes in cancer patients.
- Analysis of cancer molecular data: whole genome sequencing, RNA-Seq, DNA methylation, proteomics.

Senior Bioinformatician

April 2022 - Nov 2025

Jonsson Comprehensive Cancer Center, University of California Los Angeles, CA

- Analysis of cancer molecular data: whole genome sequencing, RNA-Seq, DNA methylation, proteomics
- Biomarker identification: among thousands of molecular features, which are most associated with cancer outcomes?
- Daily git/github use for version control and collaboration
- Supervised machine learning for predicting cancer prognosis
- Unsupervised machine learning for deriving cancer subtypes
- Supervising undegraduate and graduate students on projects
- Develop open-source R packages for sharing software with broader scientific community
- Create statistical analysis plans and power/sample size estimation for grants

Staff Biostatistician

Jan 2021 - April 2022

Edwards Lifesciences, Irvine, CA

- Develop statistical analysis plans and data monitoring for clinical trials
- Power and sample size estimation
- Clinical study reports for federal regulatory agencies
- Perform analyses with large national database (STS/ACC TVT Registry) including propensity score matching for causal inference

Postdoctoral Research Associate

Aug 2018 - Jan 2021

University of Colorado Anschutz Medical Campus, Department of Biostatistics and Informatics, Aurora, CO

- Provided statistical consulting services
- Published scientific manuscripts
- Developed grants (statistical analysis plan, power/sample size estimation)
- Supervised students and MS/MPH degree biostatisticians
- Taught BIOS 6601 Intro stats course for 75 MPH students, and created annual 6-week data science machine learning short course
- Develop new statistical methodology

Research Assistant

June 2014 - July 2018

University of Minnesota, Division of Biostatistics, Minneapolis, MN

- Developed variable selection methods for high dimensional genetic data and robust methods for estimating the genetic heritability of disease traits

Statistical Genetics Research Intern

June - Aug 2013

Dordt College, Department of Mathematics, Sioux Center, IA

- Developed methods for aggregating different statistical tests together for greater sensitivity in detecting disease-related genes; resampling-based bias corrected tests

Math and Statistics Tutor

Sept 2012 - May 2014

Winona State University, Winona, MN

Education

University of Minnesota — *Minneapolis, MN, USA*

PhD Biostatistics — *2014-2018*

- GPA: 3.7

Winona State University — *Winona, MN, USA*

B.S. Mathematics and Statistics — *2010-2014*

- GPA: 4.0 | *Summa Cum Laude*

Peer-reviewed Publications

1. **Arbet, J.**, Yamaguchi, T. N., Shiah, Y.-J., … Boutros, P. C. (2026). The landscape of prostate tumor methylation. *Cancer Discovery*, OF1–OF24. https://doi.org/10.1158/2159-8290.cd-25-0761
2. Haas, R., Margolis, M. P., Wei, A., … Geschwind, D. H. (2026). Advancing precision health discovery in a genetically diverse health system. *Cell*, *189*(9), 2533–2555.e31. https://doi.org/10.1016/j.cell.2026.03.007
3. Patel, Y., Zhu, C., Yamaguchi, T. N., … Boutros, P. C. (2026). Metapipeline-DNA: A comprehensive germline and somatic genomics nextflow pipeline. *Cell Reports Methods*, *6*(3), 101340. https://doi.org/10.1016/j.crmeth.2026.101340
4. Zeltser, N., Dang, R. M. A., Hugh-White, R., … Boutros, P. C. (2025). ApplyPolygenicScore: An r package for applying polygenic risk score models. *Genetics in Medicine Open*, *3*, 103467. https://doi.org/10.1016/j.gimo.2025.103467
5. Jones, L. W., Lavery, J. A., Tsai, B. L., … Boutros, P. C. (2025). A co-clinical trial of exercise therapy in breast cancer prevention. *Clinical Cancer Research*, *31*(16), 3377–3387. https://doi.org/10.1158/1078-0432.ccr-24-4298
6. Xu, X., Zhu, H., Hugh-White, R., … He, H. H. (2025). The landscape of N6-methyladenosine in localized primary prostate cancer. *Nature Genetics*, *57*(4), 934–948. https://doi.org/10.1038/s41588-025-02128-y
7. Tohi, Y., Sahrmann, J. M., **Arbet, J.**, … Boutros, P. C. (2025). De-escalation of monitoring in active surveillance for prostate cancer: Results from the GAP3 consortium. *European Urology Oncology*, *8*(2), 347–354. https://doi.org/10.1016/j.euo.2024.07.006
8. Lippitt, W., Carlson, N. E., **Arbet, J.**, … Kechris, K. (2024). Limitations of clustering with PCA and correlated noise. *Journal of Statistical Computation and Simulation*, *94*(10), 2291–2319. https://doi.org/10.1080/00949655.2024.2329976
9. Weiner, A. B., Agrawal, R., Wang, N. K., … Reiter, R. E. (2024). Molecular hallmarks of prostate-specific membrane antigen in treatment-naïve prostate cancer. *European Urology*, *86*(6), 579–587. https://doi.org/10.1016/j.eururo.2024.09.005
10. Ni, K., Rogowitz, E., Farahmand, A. K., … Saxon, D. R. (2024). Weight loss outcomes in a veterans affairs pharmacotherapy-based weight management clinic. *Journal of the Endocrine Society*, *8*(5). https://doi.org/10.1210/jendso/bvae042
11. Greca, A. L., Grau, L., **Arbet, J.**, … Kitahara, C. M. (2023). Anthropometric, dietary, and lifestyle factors and risk of advanced thyroid cancer: The NIH-AARP diet and health cohort study. *Clinical Endocrinology*, *99*(6), 586–597. https://doi.org/10.1111/cen.14970
12. Gamallat, Y., Choudhry, M., Li, Q., … Bismar, T. A. (2023). Serrate RNA effector molecule (SRRT) is associated with prostate cancer progression and is a predictor of poor prognosis in lethal prostate cancer. *Cancers*, *15*(10), 2867. https://doi.org/10.3390/cancers15102867
13. Chan, T. W., Dodson, J. P., **Arbet, J.**, … Xiao, X. (2022). Single-cell analysis in lung adenocarcinoma implicates RNA editing in cancer innate immunity and patient prognosis. *Cancer Research*, *83*(3), 374–385. https://doi.org/10.1158/0008-5472.can-22-1062
14. Okamoto, Y., Devoe, S., Seto, N., … Demoruelle, M. K. (2021). Association of sputum neutrophil extracellular trap subsets with IgA anti–citrullinated protein antibodies in subjects at risk for rheumatoid arthritis. *Arthritis & Rheumatology*, *74*(1), 38–48. https://doi.org/10.1002/art.41948
15. Lin, N. W., **Arbet, J.**, Mroz, M. M., … Maier, L. A. (2022). Clinical phenotyping in sarcoidosis using cluster analysis. *Respiratory Research*, *23*(1). https://doi.org/10.1186/s12931-022-01993-z
16. Grau, L., **Arbet, J.**, Ostendorf, D. M., … Creasy, S. A. (2022). Creating an algorithm to identify indices of sleep quantity and quality from a wearable armband in adults. *Sleep Science*, *15*(03), 279–287. https://doi.org/10.5935/1984-0063.20220052
17. Creasy, S. A., Ostendorf, D. M., Blankenship, J. M., … Catenacci, V. A. (2022). Effect of sleep on weight loss and adherence to diet and physical activity recommendations during an 18-month behavioral weight loss intervention.

International Journal of Obesity, *46*(8), 1510–1517. https://doi.org/10.1038/s41366-022-01141-z
18. Wood, C., **Arbet, J.**, Amura, C. R., … Anderson, J. (2021). Multicenter study evaluating nitrous oxide use for labor analgesia at high- and low-altitude institutions. *Anesthesia & Analgesia*, *134*(2), 294–302. https://doi.org/10.1213/ane.0000000000005712
19. Rosenthal, L., Lee, S., Jenkins, P., … Nodine, P. (2021). A survey of mental health in graduate nursing students during the COVID-19 pandemic. *Nurse Educator*, *46*(4), 215–220. https://doi.org/10.1097/nne.0000000000001013
20. Reed, S. M., **Arbet, J.**, & Staubli, L. (2021). Clinical nurse specialists in the united states registered with a national provider identifier. *Clinical Nurse Specialist*, *35*(3), 119–128. https://doi.org/10.1097/nur.0000000000000592
21. **Arbet, J.**, Zhuang, Y., Litkowski, E., … Kechris, K. (2021). Comparing statistical tests for differential network analysis of gene modules. *Frontiers in Genetics*, *12*. https://doi.org/10.3389/fgene.2021.630215
22. Nodine, P. M., **Arbet, J.**, Jenkins, P. A., … Hoon, S. (2021). Graduate nursing student stressors during the COVID-19 pandemic. *Journal of Professional Nursing*, *37*(4), 721–728. https://doi.org/10.1016/j.profnurs.2021.04.008
23. Thomas, E. A., Zaman, A., Cornier, M.-A., … Rynders, C. A. (2020). Later meal and sleep timing predicts higher percent body fat. *Nutrients*, *13*(1), 73. https://doi.org/10.3390/nu13010073
24. **Arbet, J.**, Brokamp, C., Meinzen-Derr, J., … Spratt, H. M. (2020). Lessons and tips for designing a machine learning study using EHR data. *Journal of Clinical and Translational Science*, *5*(1). https://doi.org/10.1017/cts.2020.513
25. Ramakrishnan, V. R., **Arbet, J.**, Mace, J. C., … Smith, T. L. (2021). Predicting olfactory loss in chronic rhinosinusitis using machine learning. *Chemical Senses*, *46*. https://doi.org/10.1093/chemse/bjab042
26. Ostendorf, D. M., Blankenship, J. M., Grau, L., … Catenacci, V. A. (2021). Predictors of long-term weight loss trajectories during a behavioral weight loss intervention: An exploratory analysis. *Obesity Science & Practice*, *7*(5), 569–582. https://doi.org/10.1002/osp4.530
27. Schmanski, A., Roberts, E., Coors, M., … Taylor, M. R. G. (2020). Research participant understanding and engagement in an institutional, self-consent biobank model. *Journal of Genetic Counseling*, *30*(1), 257–267. https://doi.org/10.1002/jgc4.1316
28. Carpenter, C. M., Frank, D. N., Williamson, K., … Kroehl, M. E. (2021). tidyMicro: A pipeline for microbiome data analysis and visualization using the tidyverse in r. *BMC Bioinformatics*, *22*(1). https://doi.org/10.1186/s12859-021-03967-2
29. **Arbet, J.**, McGue, M., & Basu, S. (2020). A robust and unified framework for estimating heritability in twin studies using generalized estimating equations. *Statistics in Medicine*, *39*(27), 3897–3913. https://doi.org/10.1002/sim.8564
30. Coleman-Minahan, K., Sheeder, J., **Arbet, J.**, …. (2020). Interest in medication and aspiration abortion training among colorado nurse practitioners, nurse midwives, and physician assistants. *Women's Health Issues*, *30*(3), 167–175. https://doi.org/10.1016/j.whi.2020.02.001
31. Gance-Cleveland, B., Linton, A., **Arbet, J.**, … Sylvain, G. (2020). Predictors of overweight and obesity in childhood cancer survivors. *Journal of Pediatric Oncology Nursing*, *37*(3), 154–162. https://doi.org/10.1177/1043454219897102
32. James-Allan, L. B., **Arbet, J.**, Teal, S. B., … Jansson, T. (2019). Insulin stimulates GLUT4 trafficking to the syncytiotrophoblast basal plasma membrane in the human placenta. *The Journal of Clinical Endocrinology & Metabolism*, *104*(9), 4225–4238. https://doi.org/10.1210/jc.2018-02778
33. Grinde, K. E., **Arbet, J.**, Green, A., … Tintle, N. (2017). Illustrating, quantifying, and correcting for bias in post-hoc analysis of gene-based rare variant tests of association. *Frontiers in Genetics*, *8*. https://doi.org/10.3389/fgene.2017.00117
34. **Arbet, J.**, McGue, M., Chatterjee, S., …. (2017). Resampling-based tests for lasso in genome-wide association studies. *BMC Genetics*, *18*(1). https://doi.org/10.1186/s12863-017-0533-3
35. Greco, B., Hainline, A., **Arbet, J.**, … Tintle, N. (2015). A general approach for combining diverse rare variant association tests provides improved robustness across a wider range of genetic architectures. *European Journal of Human Genetics*, *24*(5), 767–773. https://doi.org/10.1038/ejhg.2015.194

Awards

Postdoctoral

Staff appreciation and recognition (STAR) *2023-2024*

UCLA Jonsson Comprehensive Cancer Center

Staff appreciation and recognition (STAR) *2022-2023*

UCLA Jonsson Comprehensive Cancer Center

Student Awards

Outstanding Research Assistant Award *2017-2018*

University of Minnesota, Division of Biostatistics & Health Data Science

Interdisciplinary Biostatistics Training Grant in Genetics and Genomics *2015-2017*

National Institute of General Medical Sciences

Competitions

Midwest Undergraduate Data Analysis Competition (MUDAC) *2014*

School of Education, University of Michigan

- 3rd place out of 20 teams from 14 universities

1st place in the "predictive-modeling class competition" *2014*

Winona State University Data Mining Course

Midwest Undergraduate Data Analysis Competition (MUDAC) *2013*

School of Education, University of Michigan

- 2nd place out of 15 teams from 10 universities

Presentations

Invited Guest Speaker

The Landscape of Prostate Tumour Methylation

Prostate Cancer Foundation Precision Medicine Working Group, Virtual Meeting *2025*

- https://www.biorxiv.org/content/10.1101/2025.02.07.637178v3

Departmental

Mendelian Randomization Causal inference in Observational Studies

Statistics Working Group, Cancer Data Sciences, University of California, Los Angeles, CA *2025*

- https://jarbet.github.io/presentation-mendelian-randomization/

Batch effects: problem and solution

Boutros Lab, Cancer Data Sciences, University of California, Los Angeles, CA *2025*

- https://jarbet.github.io/presentations/2025-05-08_batch_effects_nowhite.pdf

p-value histograms

Boutros Lab, Cancer Data Sciences, University of California, Los Angeles, CA *2025*

- https://jarbet.github.io/presentation-pvalue-histograms/

Robust regression for noisy biological data

Statistics Working Group, Cancer Data Sciences, University of California, Los Angeles, CA *2024*

- https://jarbet.github.io/presentation-ordinal-regression/ordinal_regression.html

Linear regression diagnostics

Boutros Lab, Cancer Data Sciences, University of California, Los Angeles, CA *2024*

- https://jarbet.github.io/presentation-regression-diagnostics/regression_diagnostics.html

Causal inference with observational data using propensity score matching

Statistics Working Group, Cancer Data Sciences, University of California, Los Angeles, CA *2024*

- https://jarbet.github.io/presentation-causal-inference-matching/causal_inference_matching.html

Automatic feature selection and engineering with MARS

Statistics Working Group, Cancer Data Sciences, University of California, Los Angeles, CA *2023*

- https://jarbet.github.io/presentation-mars.github.io/presentation-mars.html

Introduction to Bayesian statistics

Boutros Lab, Cancer Data Sciences, University of California, Los Angeles, CA *2022*

- https://jarbet.github.io/presentations/2022-08-25_Bayesian_stats.pdf

Multiomics cancer data analysis

Boutros Lab, Cancer Data Sciences, University of California, Los Angeles, CA *2021*

- https://jarbet.github.io/presentations/2021-11-21_multiomics_cancer.pdf

Interpretable machine learning

Center for Innovative Design and Analysis, University of Colorado, Denver, CO *2021*

- https://jarbet.github.io/presentations/2020-01-21_interpretable_machine_learning.pdf

Differential Network Analysis of Gene Modules

Systems Genetics and Bioinformatics Research Lab, University of Colorado, Denver, CO *2021*

Frequentist versus Bayesian Approaches to Statistics: A Debate

Department of Biostatistics and Informatics, University of Colorado, Denver, CO *2018*

Robust Estimation of Genetic and Environmental Variance Components in Twin Studies

University of Minnesota Biostatistics Student Seminar Series, Minneapolis, MN *2017*

A General framework for Association Tests with Multivariate Traits in Large-Scale Genomics Studies

University of Minnesota Statistical Genetics Journal Club, Minneapolis, MN *2016*

Methods of Inference for Penalized Regression in High-Dimensional Genetic Association Studies

School of Public Health Research Day, University of Minnesota, Minneapolis, MN *2016*

Penalized Regression for High-Dimensional Genetic Association Testing

University of Minnesota Statistical Genetics Journal Club, Minneapolis, MN *2016*

Conferences

The Landscape of Prostate Tumour Methylation *Poster*

UCLA Health Jonsson Comprehensive Cancer Center Annual Symposium, Los Angeles, CA *2025*

Methylome-Genome Interactions Define Disease Aggression in Localized Prostate Cancer *Poster*

RECOMB/ISCB Conference on Regulatory and Systems Genomics, Los Angeles, CA *2024*

Triplet matching: propensity score matching with 3 groups *Talk*

14th Annual FDA/AdvaMed Medical Device Statistical Issues Conference (Virtual) *2022*

- https://jarbet.github.io/presentations/20220422_triplet_matching.pdf

Statistical Considerations and Methods to Utilize Real World Evidence in Medical Device Evaluation *Panel Co-organizer*

14th Annual FDA/AdvaMed Medical Device Statistical Issues Conference (Virtual) *2022*

Robust Estimation of Genetic and Environmental Variance Components in Twin Studies using Generalized Estimating Equations *Poster*

Joint Statistical Meetings (JSM), Baltimore, MD *2017*

Methods of Inference for Penalized Regression in High-Dimensional Genetic Association Studies *Poster*

Eastern North American Region International Biometric Society (ENAR), Austin, TX *2016*

What now? Post-Hoc Approaches for Gene-based Rare-Variant Tests of Association *Poster*

American Society of Human Genetics Conference (ASHG), Boston, MA *2013*

General Approaches for Combining Rare Variant Association Tests Provide Improved Power Across a Wide Range of Genetic Architectures *Poster*

American Society of Human Genetics Conference (ASHG), Boston, MA *2013*

Teaching Experience

Semester courses

BIOS 6601 – Applied Biostatistics I *Fall 2019*

University of Colorado Anschutz Medical Campus

- ***Duty***: Sole Instructor, completely redesigned all lectures, homework, quizzes, and exams
- ***Description***: Applied biostatistical methods including descriptive and statistical inference; odds ratio and relative risk, probability theory, parameter estimation, tests for comparing statistics of two or more groups, correlation, linear regression, logistic regression and survival analysis.
- ***Credits***: 3
- ***Audience***: 75 Masters of Public Health (MPH) students

Short courses

Fundamentals of Data Science Literacy *Summer 2019, Fall 2020*

University of Colorado Anschutz Medical Campus

- ***Duty***: lead organizer, developed and taught 3 machine learning lectures
- ***Description***: 6-week short course (2 hours/week) on data science with emphasis on how machine learning and data mining are used in biomedical research. Topics included: 1) Overview of Data Science; 2) Introduction of Machine Learning in Biomedical Research; 3) Clustering and Pattern Finding (unsupervised Machine Learning); 4) Powerful Predictive Models and Variable Selection (supervised Machine Learning); 5) Fundamentals of Data Visualization; 6): Reproducible Research and Team Science
- ***Audience***: 20 medical researchers, mostly non-statisticians

Introduction to Statistics *Summer 2019, 2020*

University of Colorado Anschutz Medical Campus

- ***Duty***: Instructor, developed and taught lectures on "Introduction to statistics", study design, and survey design
- ***Description***: Introductory statistics day-long short course for UCHealth nurses
- ***Audience***: 15 UCHealth Nurses

Guest Lectures

IPHY 5900 – Data Literacy in Biomedical Research *Nov 2019, Oct 2020*

University of Colorado Boulder

- ***Primary Instructor***: Dr. Celine Marie Claire D. Vetter, Integrative Physiology
- ***Lecture***: Machine Learning in Biomedical Research

BIOS 6611 – Biostatistical Methods I *Fall 2018*

University of Colorado School of Public Health

- ***Primary Instructor***: Dr. Alex Kaizer
- ***Lecture***: Linear Algebra Review

Mentoring and supervising

University of California Los Angeles

Supervising students for data analysis projects

- Yuxi Song
- Sattwik Banerjee
- Adriana Wiggins

University of Colorado, Department of Biostatistics and Informatics

Supervising MPH/MS Statisticians

- Laura Grau (MPH), Nov 2018 – Jan 2021
  - I supervised Laura on several projects which led to publications.

Supervising students for data analysis projects

- Yaxu Zhuang (PhD), Jan 2019 – Aug 2020
- Rachel Weber (MS), Dec 2018 – April 2019

Thesis or Dissertation Committee Member (graduating year)

- Charlie Carpenter, Biostatistics MS thesis committee, Biostatistics (2020)
- Rachel Blumhagen, Biostatistics PhD dissertation committee (2021)

Professional Service

Grant and Protocol Reviews

- Scientific Advisory & Review Committee (SARC), Colorado Clinical and Translational Sciences Institute, University of Colorado – April 2019 to Jan 2021

Journal Reviewer

- Cancer Research
- Journal of the American Heart Association (JAHA)
- Genetic Epidemiology
- PeerJ
- Biometrics
- Journal of Open Source Software (JOSS)

Committees/Other

Jonsson Comprehensive Cancer Center, University of California Los Angeles

- Chair of Statistics Working Group: April 2023 - Present

Department of Biostatistics and Informatics, University of Colorado

- Student Admissions Committee, Fall 2018 – Spring 2019
- Volunteer Judge for CIDA Hackathon (24 hour data analysis competition), Jan. 2020

Biostatistics, Epidemiology, & Research Design Core of the Colorado Clinical and Translational Sciences Institute, University of Colorado

- Education Committee, Fall 2018 – Jan 2020

Software Skills

Proficient

R

- Over **15 years** of experience with R for data analysis
- I enjoy **developing R packages** to easily share documented code with colleagues and the broader research community. Examples:
  - `https://uclahs-cds.github.io/package-PrCaMethy/`
  - `https://github.com/arbet003/discoMod`
- **Parallel programming** in R using multiple cpus

git/github

- Version control
- Used daily
- Collaborating with others on projects
- Review colleagues' code
- Assign issues to team members

Quarto and Rmarkdown

- Reproducible research, manuscripts and reports
- Reproducible HTML presentations. Examples: `https://jarbet.github.io/presentations`
- Websites: `https://jarbet.github.io/`
- Resume: this resume was made using Rmarkdown with the vitae R package

Bayesian statistical programming

- Proficient with BUGS/JAGS; basic skills with STAN

Basic skills

- Linux command line and bash scripts
  - Daily use of a high-performance computing cluster at UCLA
- Databricks
- Microsoft Office
- SAS